

**JKBOSE 2026**

**CLASS 11<sup>TH</sup>**

**STATISTICS**

**MATHS**

**LECTURE-4**

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# TOPICS TO BE COVERED

1

$\sigma^2$  and  $\sigma$  by Short Cut Method

2

Exercise Discussion 13.2

$$\bar{x} = \frac{\sum f_i x_i}{N}$$

$$\overline{x^2} = \frac{\sum f_i x_i^2}{N}$$

$$\sigma = \sqrt{\overline{x^2} - (\bar{x})^2}$$



**JKBOSE  
WALLAH**

# 7 DAYS BOARD CHALLENGE



**COMPLETE YOUR BOARD SYLLABUS**



## NCERT REVISION

Strengthen Concepts.  
Build Clarity.



## MCQ PRACTICE

Practice More.  
Score More.



## FORMULA SHEET

Quick Revision.  
Smart Recall.

**DAY 1**



### NCERT REVISION

Read. Understand.  
Revise.

**DAY 2**



### NCERT REVISION

Revise. Remember.  
Strengthen.

**DAY 3**



### NCERT REVISION

Concepts Clear.  
Confidence Strong.

**DAY 4**



### MCQ PRACTICE

Practice MCQs.  
Improve Accuracy.

**DAY 5**



### MCQ PRACTICE

More Practice.  
Better Scores.

**DAY 6**



### FORMULA SHEET

Revise Formulas.  
Lock Important  
Points.

**DAY 7**



### FORMULA SHEET

Quick Revision.  
Be Exam Ready.



**SET A TARGET  
EVERY DAY**



**COMPLETE  
YOUR TARGET**



**SHARE DONE  
IN THE GROUP**



**STAY CONSISTENT  
STAY AHEAD**



## 7 DAYS. ONE GOAL. COMPLETE BOARD REVISION.

**SMALL DAILY EFFORTS | BIG RESULTS ON EXAM DAY**

py Q papers

Subjective Questions  
+ Objectives

# $\sigma^2$ and $\sigma$ by Short Cut Method

$$\sigma_x = h\sigma_y$$

Suppose that the observations are given by;

|       |       |       |       |         |       |
|-------|-------|-------|-------|---------|-------|
| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $\dots$ | $x_n$ |
| $f_1$ | $f_2$ | $f_3$ | $f_4$ | $\dots$ | $f_n$ |

let  $A$  be the assumed mean of the given data & defined the new - observations as follows:

$$y_i = \frac{x_i - A}{h}$$

$$\Rightarrow x_i = A + hy_i \text{ --- (1), } \forall i = 1, 2, 3, \dots, n$$

$$\text{Here } \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N}$$

$$\Rightarrow \bar{x} = \frac{\sum_{i=1}^n f_i (A + hy_i)}{N} = \frac{\sum_{i=1}^n f_i A}{N} + \frac{\sum_{i=1}^n f_i h y_i}{N}$$

$$\Rightarrow \bar{x} = A \frac{\sum_{i=1}^n f_i}{N} + h \frac{\sum_{i=1}^n f_i y_i}{N} = \frac{AN}{N} + h \bar{y}$$

$$\Rightarrow \bar{x} = A + h\bar{y} \text{ --- (2)}$$

$$\text{Also, } \sigma_x^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{N} = \frac{\sum_{i=1}^n f_i (A + hy_i - A - h\bar{y})^2}{N}$$

$$\Rightarrow \sigma_x^2 = \frac{\sum_{i=1}^n f_i (hy_i - h\bar{y})^2}{N} = h^2 \frac{\sum_{i=1}^n f_i (y_i - \bar{y})^2}{N}$$

$$\Rightarrow \sigma_x^2 = h^2 \sigma_y^2 \text{ Hence the result.}$$

## EXAMPLE



$$y = -\frac{15}{50} = -0.3$$

Calculate mean, variance and standard deviation for the following distribution.

| Classes   | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 | 80-90 | 90-100 |
|-----------|-------|-------|-------|-------|-------|-------|--------|
| Frequency | 3     | 7     | 12    | 15    | 8     | 3     | 2      |

(Sol) Suppose that  $A = 65$  be the assumed mean & define the new observations as follows;

$$y_i = \frac{x_i - A}{h} = \frac{x_i - 65}{10}$$

Therefore By Short Cut Method, we get

$$\bar{x} = A + h\bar{y}$$

$$\Rightarrow \bar{x} = 65 + 10 \times \frac{3}{50} = 65 - 3$$

$$\Rightarrow \boxed{\bar{x} = 62}$$

$$\text{Also, } \sigma_x^2 = h^2 \sigma_y^2$$

$$\Rightarrow \sigma_x^2 = (10)^2 \times \left( (\overline{y^2}) - (\bar{y})^2 \right)$$

$$\Rightarrow \sigma_x^2 = 100 \left( \frac{105}{50} - (-0.3)^2 \right)$$

$$\Rightarrow \sigma_x^2 = 100 \left( \frac{105}{50} - 0.09 \right)$$

$$\Rightarrow \sigma_x^2 = 2 \times 105 - 100 \times 0.09$$

$$\Rightarrow \sigma_x^2 = 210 - 9$$

$$\Rightarrow \sigma_x^2 = 201$$

$$\sigma_x = \sqrt{201}$$

Therefore Required

$$\text{Mean} = 62$$

$$\text{Variance} = 201$$

$$\text{Standard deviation} = \sqrt{201}$$

Hence the result.

| Class  | frequency<br>( $f_i$ ) | class mark<br>( $x_i$ ) | $y_i = \frac{x_i - 65}{10}$ | $f_i y_i$ | $y_i^2$ | $f_i y_i^2$ |
|--------|------------------------|-------------------------|-----------------------------|-----------|---------|-------------|
| 30-40  | 3                      | 35                      | -3                          | -9        | 9       | 27          |
| 40-50  | 7                      | 45                      | -2                          | -14       | 4       | 28          |
| 50-60  | 12                     | 55                      | -1                          | -12       | 1       | 12          |
| 60-70  | 15                     | 65                      | 0                           | 0         | 0       | 0           |
| 70-80  | 8                      | 75                      | 1                           | 8         | 1       | 8           |
| 80-90  | 3                      | 85                      | 2                           | 6         | 4       | 12          |
| 90-100 | 2                      | 95                      | 3                           | 6         | 9       | 18          |
|        | 50                     |                         |                             | -15       |         | 105         |

$$y^2 = \frac{\sum f_i y_i^2}{N}$$

# EXERCISE – 13.2

## QUESTION

Find the mean and variance for each of the data in Exercises 1 to 5.

6, 7, 10, 12, 13, 4, 8, 12 *H/W*

## QUESTION



Find the mean and variance for each of the data in Exercises 1 to 5.

First  $n$  natural numbers

$$\sigma = \frac{n^2 - 1}{12}$$

H/W

$$\frac{n(n+1)}{2} = n+1$$

$$\bar{x} = \frac{1+2+3+\dots+n}{n}$$

$$\Rightarrow \bar{x} = \frac{n(n+1)}{2} = \frac{n+1}{2}$$

## QUESTION

Find the mean and variance for each of the data in Exercises 1 to 5.

First 10 multiples of 3

$$\begin{aligned}
 \text{(Sol)} \quad \bar{x} &= \frac{3 \times 1 + 3 \times 2 + 3 \times 3 + \dots + 3 \times 10}{10} \\
 &= \frac{3}{10} (1 + 2 + 3 + \dots + 10) \\
 &= \frac{3}{\cancel{10}} \frac{(\cancel{10} \times 11)}{2} = 3 \times \frac{11}{2}
 \end{aligned}$$

# QUESTION

Find the mean and variance for each of the data in Exercises 1 to 5.

|       |   |    |    |    |    |    |    |
|-------|---|----|----|----|----|----|----|
| $x_i$ | 6 | 10 | 14 | 18 | 24 | 28 | 30 |
| $f_i$ | 2 | 4  | 7  | 12 | 8  | 4  | 3  |

H/W

## QUESTION

Find the mean and variance for each of the data in Exercises 1 to 5.

|       |    |    |    |    |     |     |     |
|-------|----|----|----|----|-----|-----|-----|
| $x_i$ | 92 | 93 | 97 | 98 | 102 | 104 | 109 |
| $f_i$ | 3  | 2  | 3  | 2  | 6   | 3   | 3   |

H/w

## QUESTION

Find the mean and standard deviation using short-cut method

|       |    |    |    |    |    |    |    |    |    |
|-------|----|----|----|----|----|----|----|----|----|
| $x_i$ | 60 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 |
| $f_i$ | 2  | 1  | 12 | 29 | 25 | 12 | 10 | 4  | 5  |

(Sol) Suppose that  $A = 64$  be the assumed mean & define the new observations as follows:

$$y_i = \frac{x_i - A}{h} = x_i - 64$$

Therefore by short cut Method, we get

$$\bar{x} = A + h\bar{y}$$

$$\Rightarrow \bar{x} = A + \bar{y} = 64 + \left(\frac{0}{100}\right)$$

$$\Rightarrow \boxed{\bar{x} = 64}$$

$$\text{Also } \sigma_x^2 = h^2 \sigma_y^2 = 1^2 \times \left( \overline{y^2} - (\bar{y})^2 \right)$$

$$\Rightarrow \sigma_x^2 = \frac{286}{100} - 0^2$$

$$\Rightarrow \sigma_x^2 = 2.86$$

$$\therefore \text{Standard deviation} = \sqrt{2.82} = 1.68$$

Hence the result.

| $x_i$ | $f_i$ | $y_i = x_i - 64$ | $f_i y_i$ | $y_i^2$ | $f_i y_i^2$ |
|-------|-------|------------------|-----------|---------|-------------|
| 60    | 2     | -4               | -8        | 16      | 32          |
| 61    | 1     | -3               | -3        | 9       | 9           |
| 62    | 12    | -2               | -24       | 4       | 48          |
| 63    | 29    | -1               | -29       | 1       | 29          |
| 64    | 25    | 0                | 0         | 0       | 0           |
| 65    | 12    | 1                | 12        | 1       | 12          |
| 66    | 10    | 2                | 20        | 4       | 40          |
| 67    | 4     | 3                | 12        | 9       | 36          |
| 68    | 5     | 4                | 20        | 16      | 80          |
|       | 100   |                  | 0         |         | 286         |

## QUESTION

Find the mean and variance for the following frequency distributions in Exercises 7 to 8.

| Classes   | 0–30 | 30–60 | 60–90 | 90–120 | 120–150 | 150–180 | 180–210 |
|-----------|------|-------|-------|--------|---------|---------|---------|
| Frequency | 3    | 3     | 5     | 10     | 3       | 5       | 2       |

H/w

## QUESTION

Find the mean and variance for the following frequency distributions in Exercises 7 to 8.

| Classes   | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 |
|-----------|------|-------|-------|-------|-------|
| Frequency | 5    | 8     | 15    | 16    | 6     |

H/w

## QUESTION



Find the mean, variance and standard deviation using short-cut method.

| Height in cms   | 70-75 | 75-80 | 80-85 | 85-90 | 90-95 | 95-100 | 100-105 | 105-110 | 110-115 |
|-----------------|-------|-------|-------|-------|-------|--------|---------|---------|---------|
| No. of children | 3     | 4     | 7     | 7     | 15    | 9      | 6       | 6       | 3       |

H/W

## QUESTION

Find the mean, variance and standard deviation using short-cut method

|                | 32.5-36.5 | 36.5-40.5 | 40.5-44.5 | 44.5-48.5 | 48.5-52.5 |
|----------------|-----------|-----------|-----------|-----------|-----------|
| Diameters      | 33-36     | 37-40     | 41-44     | 45-48     | 49-52     |
| No. of circles | 15        | 17        | 21        | 22        | 25        |

Calculate the standard deviation and mean diameter of the circles.

**HINT:** [First make the data continuous by making the classes as 32.5-36.5, 36.5-40.5, 40.5-44.5, 44.5-48.5, 48.5-52.5 and then proceed.]

H/W

**Thank**  
**You**